

MLR - Exercise 12

Naive bayes classifier, Gaussian mixture model

Bayesian classifiers

Given measured data $\mathbf{x}$ and the class prediction $y \in \{y_1, \dots, y_K\}$. We want to obtain the posterior probability of the class:

$$p(y_k | \mathbf{x})$$

This conditional probability for k-class can be derived from observed data.

This probability can be derived from observed data as frequency of samples with k-class.

We apply a Bayesian rule:

$$p(y_k | \mathbf{x}) = \frac{p(\mathbf{x} | y_k) p(y_k)}{p(\mathbf{x})} = \frac{p(x_1, x_2 \dots x_d | y_k) p(y_k)}{\sum_k p(x_1, x_2 \dots x_d | y_k) p(y_k)}$$

and we are most interested in most probable class:

$$y_k^* = \operatorname{argmax}_{y_k} p(\mathbf{x} | y_k) p(y_k)$$

Sum over K discrete classes - it is possible.

We can select from set $\mathbf{x}$ only samples with k -th class and fit parameters of probability distribution $p_{\theta}(x_1, \dots, x_d | y_k)$ by e.g. ML estimation.

However, this distribution is d -dimensional and therefore fitting can be difficult.

Naive Bayes classifiers

$$y_k^* = \operatorname{argmax}_{y_k} p(x_1, \dots, x_d | y_k) \cdot p(y_k)$$

Assume that the features are mutually independent, conditional on given category y_k

$$y_k^* = \operatorname{argmax}_{y_k} p(x_1, \dots, x_d | y_k) \cdot p(y_k) = \operatorname{argmax}_{y_k} p(y_k) \cdot \prod_{i=1}^d p_{\theta}(x_i | y_k)$$

The task is to find distributions $p(x_i | y_k)$ described by parameter vector $\theta_{i,k}$ for i -th feature conditional by k -th class, thus we obtain $k \times d$ distributions.

The parameters of this distribution can be found with probability model based on observed data as follows.

We have observed set of n samples for i -th feature $[f_1, f_2, \dots, f_n]$. Then we estimate parameter $\theta_{i,k}$ for set of samples $\mathbf{f}_{i,k}$ corresponding to k -class. We are fitting distribution only to samples corresponding to k -class, because $p(x_i | y_k)$ is conditioned by y_k .

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta_{i,k}} p(\theta_{i,k} | \mathbf{f}_{i,k}) = \operatorname{argmax}_{\theta_{i,k}} \frac{p(\mathbf{f}_{i,k} | \theta_{i,k}) \cdot p(\theta_{i,k})}{p(\mathbf{f}_{i,k})} \quad \mathbf{f}_{i,k} = \mathbf{f}(x_i | y_k)$$

Naive Bayes - example of Bernoulli and Gaussian features

$$p(y_k|\mathbf{x}) = \frac{p(\mathbf{x}|y_k)p(y_k)}{p(\mathbf{x})} = \frac{p(x_1, x_2|y_k)p(y_k)}{\sum_k p(x_1, x_2|y_k)p(y_k)} = \frac{p_{\mu,\sigma}(x_1|y_k)p_{\theta}(x_2|y_k)p(y_k)}{\sum_k p_{\mu,\sigma}(x_1|y_k)p_{\theta}(x_2|y_k)p(y_k)}$$

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$$p_{\mu,\sigma}(f_{1,k}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(f_{1,k}-\mu)^2}{2\sigma^2}\right)$$

$$\hat{\mu}_{MLE} = \sum_{i=1}^n \frac{f_{1,k}^i}{n} = \overline{f_{1,k}}$$

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (f_{1,k}^i - \overline{f_{1,k}})^2$$

$$p_{\theta}(f_{2,k}) = \theta^{f_{2,k}}(1-\theta)^{1-f_{2,k}}$$

$$\hat{\theta}_{MLE} = \sum_{i=1}^n \frac{f_{2,k}^i}{n} = \overline{f_{2,k}}$$

$$\hat{p}(y_1) = \frac{\text{\# of samples of class } y_1}{\text{\# samples}}$$

Logistic regression

Now, we will use this sigmoid function to transform our linear model:

$$\theta = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

Bernoulli probability distribution: $f(x|\theta) = \theta^x(1 - \theta)^{1-x}$

We can now look on our data in a different way: We can interpret labels as a Bernoulli random variable given the model described by sigma function.

Now, we define likelihood as a “probability” that observed data x come from specific model (Bernoulli) with parameters $\mathbf{w}$. This likelihood function for my independent and identically distributed random variable has the following form:

$$\mathcal{L}(\mathbf{w}|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_i^n \sigma(\mathbf{w}^T \mathbf{x}_i)^{y_i} \left(1 - \sigma(\mathbf{w}^T \mathbf{x}_i)\right)^{1-y_i}$$

Solution

It turns out that maximizing likelihood function can be difficult due to multiplications. Better way is to maximize Log of likelihood function (this is binary cross-entropy):

$$\mathcal{L}_{log} = \log \mathcal{L}(\mathbf{w} |, \mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \sum_i y_i \log(\sigma(\mathbf{w}^T \mathbf{x}_i)) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i))$$

We have to maximize this function with respect to $\mathbf{w}$. So, we must take the derivative of $\mathcal{L}_{log}$

$$\frac{d \mathcal{L}_{log}(\mathbf{w} |, \mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n)}{d w^{(j)}} = \sum_i (y_i - \sigma(\mathbf{w}^T \mathbf{x}_i)) x_i^{(j)}$$

And finally, we can write the gradient ascent optimization:

$$w_{new}^{(j)} = w_{old}^{(j)} + \mu \sum_i (y_i - \sigma(\mathbf{w}^T \mathbf{x}_i)) x_i^{(j)}$$

Interpretation

Odds = $\frac{p}{1-p}$ (e.g. $\frac{0.2}{0.8}$ for 20% probability)

Log-odds - weights values

- similar to interpretation to weights in linear regression
- large positive -> increase of this feature lead to larger probability
- large negative -> increase of this feature leads to smaller probability
- close to 0 -> unimportant feature

Odds ratio - e to power of weights values $OR_j = e^{w_j}$

- if corresponding feature increases by 1, odds are multiplied by the odds ratio

e.g. :

1.42 -> increase by 42% (original odds $\frac{0.1}{0.9}$ (probability 10%), new odds $\frac{0.1}{0.9} 1.42$)

0.42 -> decrease to 42% (original odds $\frac{0.1}{0.9}$ (probability 10%), new odds $\frac{0.1}{0.9} 0.42$)

Multivariate Gaussian function

d-dimensional Gaussian function:

$$f(\mathbf{x}) = (2\pi)^{-\frac{d}{2}} |\mathbf{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

$$\begin{aligned} \boldsymbol{\mu} &= (\mu_1, \mu_2, \dots, \mu_d)^T \\ \mathbf{x} &= (x_1, x_2, \dots, x_d)^T \end{aligned} \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_1\sigma_d \\ \vdots & \ddots & \vdots \\ \sigma_d\sigma_1 & \cdots & \sigma_d^2 \end{bmatrix}$$

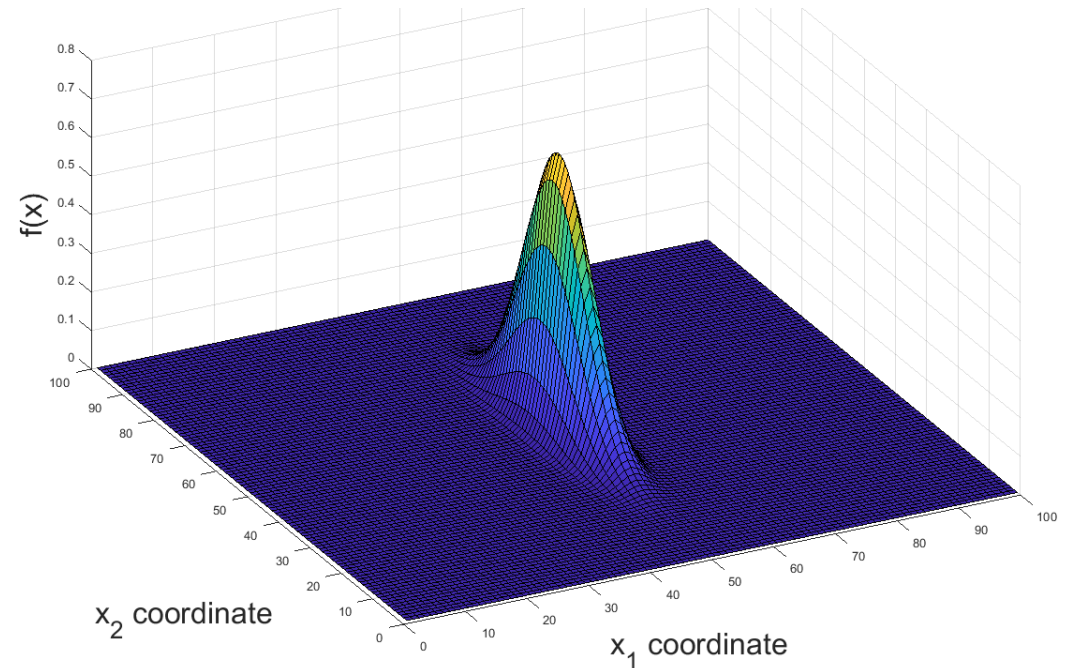
for 2-dimensional case:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_1\sigma_2 \\ \sigma_2\sigma_1 & \sigma_2^2 \end{bmatrix}$$

Estimation using MLE:

$$\boldsymbol{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i = \bar{\mathbf{x}}$$

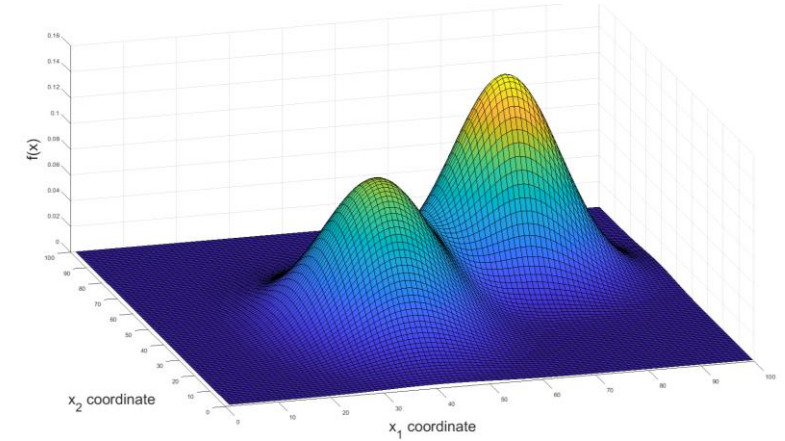
$$\mathbf{\Sigma}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu}_{\text{MLE}})^T (\mathbf{x}_i - \boldsymbol{\mu}_{\text{MLE}})$$



Gaussian mixture model

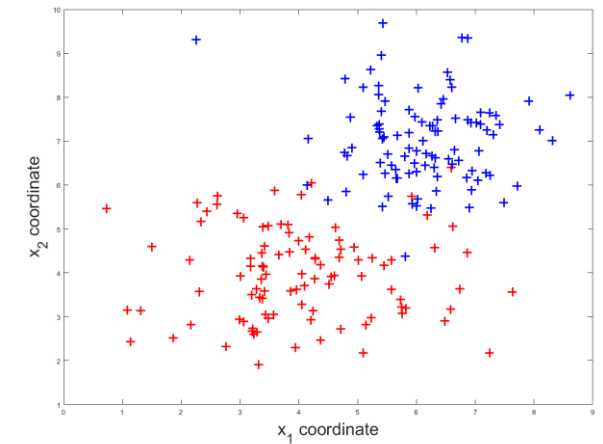
Mixture model

$$f(x; a_1, a_2, \dots, a_K, w_1, \dots, w_K) = \sum_{k=1}^K w_k p(x; a_k) \quad \sum_{k=1}^K w_k = 1$$



Multivariate Gaussian mixture model:

$$f(x; \mu_1, \dots, \mu_K, \Sigma_1, \dots, \Sigma_K, w_1, \dots, w_K) = \sum_{k=1}^K w_k p(x; \mu_k, \Sigma_k)$$



Expectation – maximization (EM) for GMM

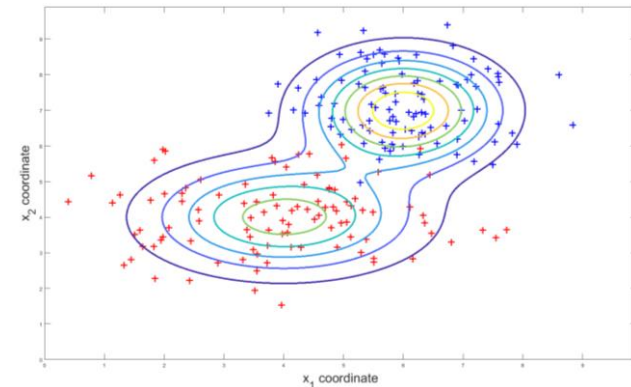
E – step – evaluate likelihood for each point x_i and each Gaussian k - $p(x_i; \theta)$ and normalize to get probability of this point to be from class k :

$$q(x_i; \theta_k) = \frac{w_k p(x_i; \theta_k)}{\sum_{j=1}^K w_j p(x_i; \theta_j)}$$

$$p(x_i; \theta_k) = (2\pi)^{-\frac{d}{2}} |\Sigma_k|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x_i - \mu_k)^T \Sigma_k^{-1}(x_i - \mu_k)\right)$$
$$\theta_k = [\mu_k, \Sigma_k]$$

M – step – get MLE estimate for each Gaussian – points are weighted by probability of points to be from class k :

$$w_k = \frac{1}{N} \sum_{i=1}^N q(x_i; \theta_k), \quad \mu_k = \frac{\sum_{i=1}^N q(x_i; \theta_k) x_i}{\sum_{i=1}^N q(x_i; \theta_k)}, \quad \Sigma_k = \frac{\sum_{i=1}^N q(x_i; \theta_k) (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^N q(x_i; \theta_k)}$$



Gaussian mixture model estimation by EM

Algorithm for case of two 2D Gaussian (d=2, K=2)

1. Initialization of $\mu_1, \mu_2, \Sigma_1, \Sigma_2, w_1, w_2$
2. calculate normalized likelihoods $q(x_i; \Theta_1)$ and $q(x_i; \Theta_2)$ for all points (E-step)

$$q(x_i; \Theta_k) = \frac{w_k p(x_i; \Theta_k)}{\sum_{j=1}^K w_j p(x_i; \Theta_j)} \quad p(x_i; \Theta_k) = (2\pi)^{-\frac{d}{2}} |\Sigma_k|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x_i - \mu_k)^T \Sigma_k^{-1} (x_i - \mu_k)\right)$$

3. calculate w_1 and w_2 as mean of $q(x_i; \Theta_k)$ over all points $w_k = \frac{1}{N} \sum_{i=1}^N q(x_i; \Theta_k)$, and re-estimate parameters μ_1, μ_2, Σ_1 and Σ_2 (M - step)

$$\mu_k = \frac{\sum_{i=1}^N q(x_i; \Theta_k) x_i}{\sum_{i=1}^N q(x_i; \Theta_k)}, \quad \Sigma_k = \frac{\sum_{i=1}^N q(x_i; \Theta_k) (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^N q(x_i; \Theta_k)}$$

4. repeat step 2. - 3. until stopping criteria are not met

Stopping criterion:

- Maximal number of iterations
- Change of parameters is under limit $|\mu^{i+1} - \mu^i| < \epsilon$

